

ROOTMESH · REQUIREMENTS

RootMesh requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/rootmesh		
Source	docs/03-requirements.md @ f37ca6b		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status for each
0.1	2026-09-24	Amish Chadha	Initial scaffold

RootMesh requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with growers, and will be checked by calculation at TRL 3 and revised after user input (see RMS-PRB-001). The status column gives the concept's position against each target from the first-order numbers in RMS-PRC-001. Nothing has been built or measured, so "met" means met on paper or by a datasheet.

Reference case: one stake in a loam field, sensing at 150 and 300 mm, reporting every 20 min to an indoor gateway about 1 km away, in a region with 3 peak sun hours per day in the worst month.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Measure volumetric water content at two root-zone depths	150 mm and 300 mm below grade (± 25 mm); 5 to 50 % VWC; ± 3 % VWC after a site-specific calibration	Calibration method note; later gravimetric calibration in two soils	Unverified, at risk: depends on probe contact and calibration
R2	Measure root-zone soil temperature	± 0.5 °C from -10 to 60 °C at about 225 mm depth	DS18B20 datasheet	Met by datasheet
R3	Measure bulk soil electrical conductivity	0 to 4 dS/m; ± 10 % or ± 0.1 dS/m, whichever is larger, after calibration in standard solutions; reported at 25 °C	Circuit calculation; later bench test in KCl solutions and soil	Unverified
R4	Report often enough for daily irrigation decisions	Every 20 min or shorter by default, adjustable from 10 to 60 min; uplink airtime 30 s or less per day at the design spreading factor (SF10)	Airtime calculation	Met at 20 min (about 27 s per day at SF10). A 15 min interval is not possible at SF10 within 30 s per day
R5	Reach the gateway	90 % or more of uplinks received at 1 km from an indoor gateway, stake antenna about 0.2 m above grade, open field with a mature crop	Link budget; later field range walk	At risk, likely not met with an indoor gateway; see RMS-PRC-001
R6	Run on sun alone	Energy-neutral with 1 peak sun hour per day on the panel; 90 days or more of reporting with no harvest at all	Energy budget	Met on estimate: about 70 times daily use harvested at 3 sun hours; about 240 days on the cell alone
R7	Safe cell and charging	LiFePO4 chemistry, 2 Wh or less per stake; charging blocked below 0 °C and above 45 °C cell temperature; cell fused	Design review; charger datasheet	Met by design (about 1.9 Wh, NTC cut-off, PTC fuse)
R8	Keep water out	Head IP67; buried section IP68 at 0.5 m for a growing season; UV-stable exposed parts	Design review; later immersion and spray test	Unverified
R9	Operate across field temperatures	-10 to 60 °C at the head, -5 to 40 °C at the cell	Component datasheets; thermal estimate	Met by datasheets; head temperature in full sun is an estimate
R10	Install and remove by hand	One person with a 50 mm hand auger installs a stake in 15 min or less, with sensing depths repeatable to ± 25 mm; removal by hand in 5 min or less	Installation sequence review; later timed trial	Unverified
R11	Survive field operations	Head and antenna 300 mm or less above grade; high-visibility marker 1 m or more above grade	Model check	Met by design (antenna tip about 270 mm, flag about 1 m)
R12	Affordable	One stake \$60 or less in parts; pilot set of three stakes and one gateway \$300 or less	Priced BOM (bom/ bom.csv)	Met on indicative prices (about \$52 per stake, about \$246 per pilot set)

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R13	Grower owns the data	Data readable without a paid subscription; CSV export; open dashboard	Design review	Partly met: free with The Things Network; a fully local server needs a different gateway (open question)
R14	Last in the ground	Probes and electrodes last 2 growing seasons (about 12 months buried) without replacement	Materials review; later soak test	Unverified, at risk: hobby probes corrode quickly unless sealed

■ Assumptions

- Soil water content, not soil water potential, is the quantity measured. Converting it to plant-available water needs field capacity and wilting point for the soil, set during calibration (FAO-56 method).
- EC is measured as bulk soil EC. Pore-water or saturated-paste EC, which crop salinity thresholds use, differ and need a moisture-dependent conversion.
- LoRaWAN uplink payload of about 11 bytes (two moisture counts, EC, temperature, cell voltage, status), 24 bytes on air with LoRaWAN overhead.
- SF10 at 125 kHz is the design case because it is the slowest uplink data rate allowed in the US915 plan and a reasonable margin case in EU868.
- The Things Network fair use policy (30 s of uplink airtime per node per day) applies when the community network is used.
- Worst-month solar resource of 3 peak sun hours on the panel before shading and dust losses.